ESA 321
Aerospace Structures

Lecture 4 – Truss Element

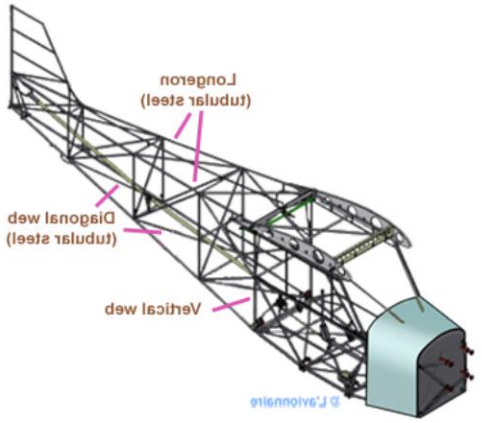
Objectives

- **To apply FEM using truss element for solving structural problems.**

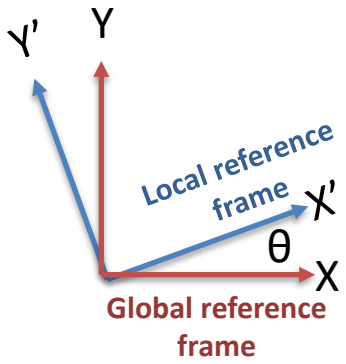
Usage

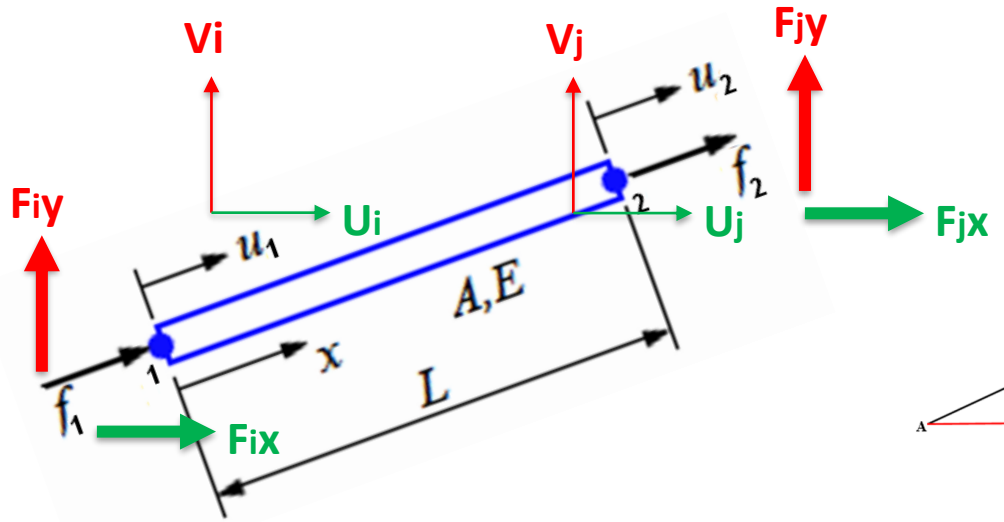
- Truss elements are used in two and three dimensions to model slender, line-like structures that support loading only along the axis or the centreline of the element.

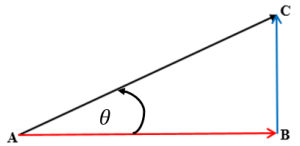












Each node has 2 DOF

Truss element formulation

$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix}$$

$$\mathbf{k} = \frac{AE}{L}\begin{bmatrix} C^2 & CS & -C^2 & -CS \\ CS & S^2 & -CS & -S^2 \\ -C^2 & -CS & C^2 & CS \\ -CS & -S^2 & CS & S^2 \end{bmatrix}$$

Where

$$l = \cos\theta = \frac{X_j - X_i}{L}$$

$$m = \sin\theta = \frac{Y_j - Y_i}{L}$$

$$C = \cos\theta$$

$$S = \sin\theta$$

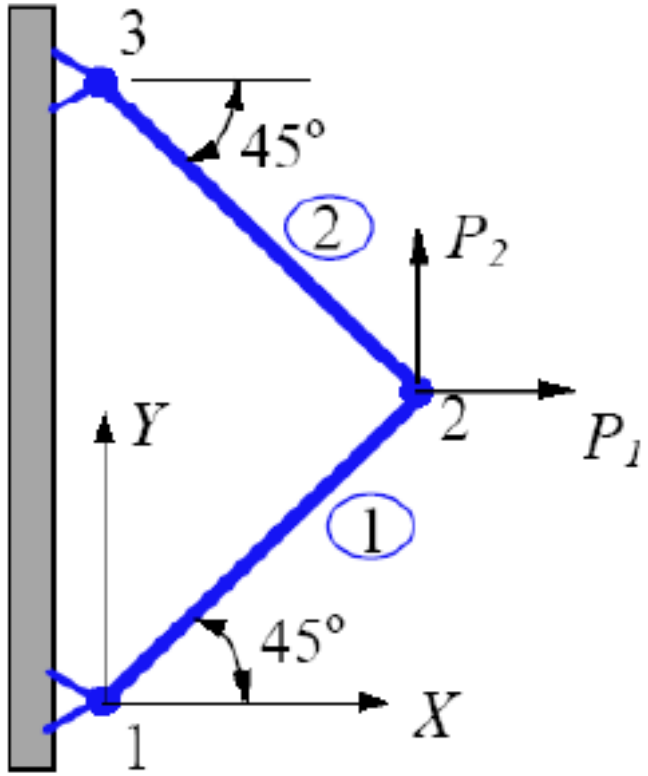
2D - Truss Element

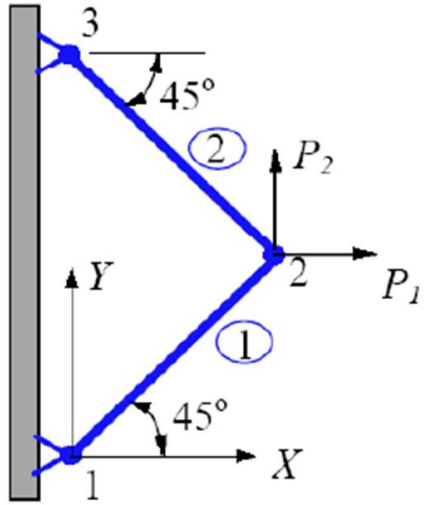
- **Properties**
  - Ideal truss members are **connected only at their ends**.
  - Ideal truss members are connected by **frictionless pins (no moments)**
  - The truss structure is **loaded only at the pins**
  - *Deformation (dof) in **horizontal** (u) and **vertical** (v)*
  - **Weights** of the members are neglected

Example

A simple plane truss is made of two identical bars (with *E, A, and L), and loaded as shown in the* figure. Find

1) displacement of node 2;

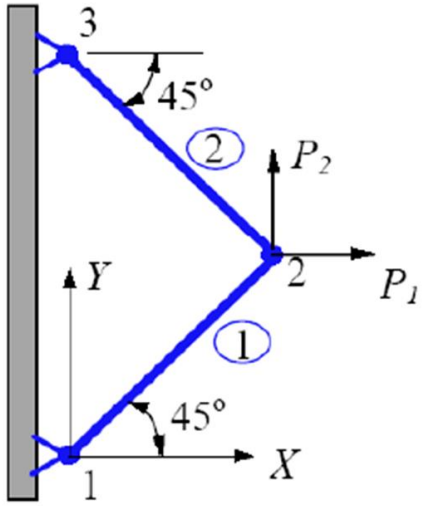




$$\begin{matrix} & u_i & v_i & u_j & v_j \\ \frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} & \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i\,x \\ F_i\,y \\ F_j\,x \\ F_j\,y \end{Bmatrix} \end{matrix}$$

$l = \cos\theta$:

$m = \sin\theta$

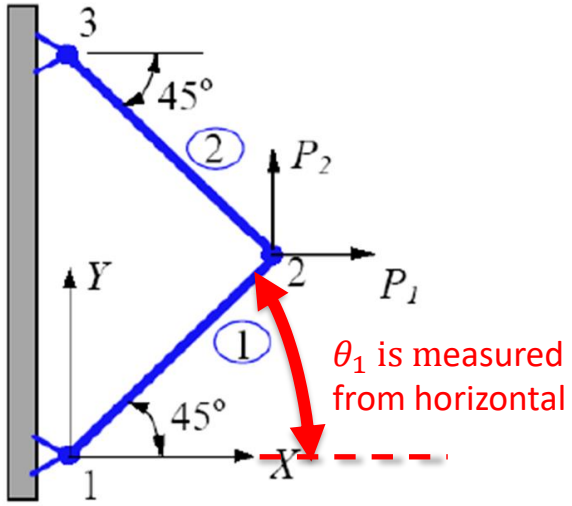


$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix} \qquad l = \cos\theta \qquad m = \sin\theta$$

Element 1 $u_1 \quad v_1 \quad u_2 \quad v_2$

$F_1 x \quad F_1 y \quad F_2 x \quad F_2 y$

$$\frac{EA}{L}\begin{bmatrix} & & & \\ & \textbf{Stiffness matrix} & & \\ & & & \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{bmatrix} F_1 x \\ F_1 y \\ F_2 x \\ F_2 y \end{bmatrix}$$



$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix}$$

$l = \cos\theta$

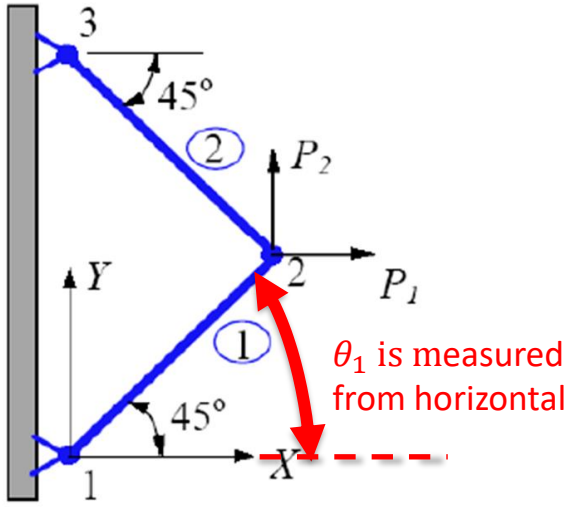
$m = \sin\theta$

Element 1 $u_1 \quad v_1 \quad u_2 \quad v_2$

$F_1 x \quad F_1 y \quad F_2 x \quad F_2 y$

$$u_1 \quad v_1 \quad u_2 \quad v_2$$

$$\frac{EA}{L}\left[\quad \textbf{Stiffness matrix} \quad\right]\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{bmatrix} F_1 x \\ F_1 y \\ F_2 x \\ F_2 y \end{bmatrix}$$



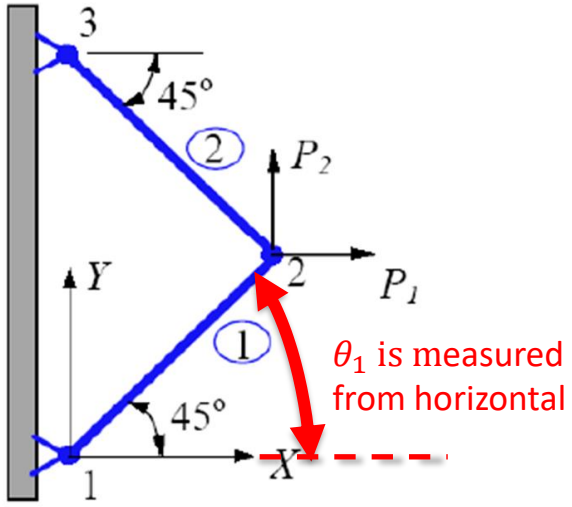
$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix} \qquad l = \cos\theta \qquad m = \sin\theta$$

Element 1 $u_1 \quad v_1 \quad u_2 \quad v_2$

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$\theta_1 = 45^o$



$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix}$$

$l = \cos\theta$

$m = \sin\theta$

Element 1 $u_1 \quad v_1 \quad u_2 \quad v_2$

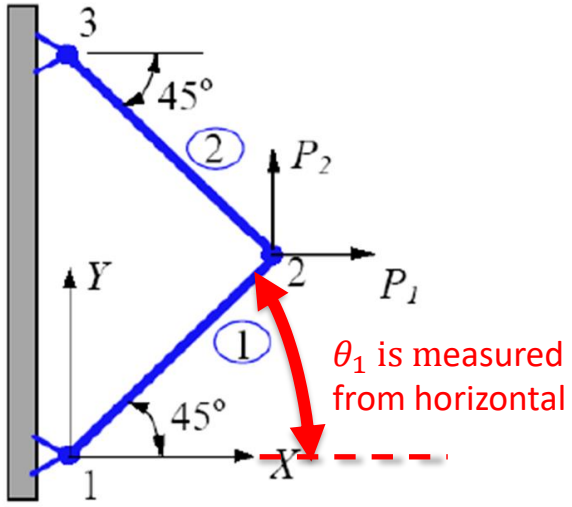
$F_1 x \quad F_1 y \quad F_2 x \quad F_2 y$

$$\frac{EA}{L}\begin{bmatrix} u_1 & v_1 & u_2 & v_2 \\ & \textbf{Stiffness matrix} & & \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{bmatrix} F_1 x \\ F_1 y \\ F_2 x \\ F_2 y \end{bmatrix}$$

$\theta_1 = 45^o$

$l =$

$m =$



$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix}$$

$l = \cos\theta$

$m = \sin\theta$

Element 1 $u_1 \quad v_1 \quad u_2 \quad v_2$

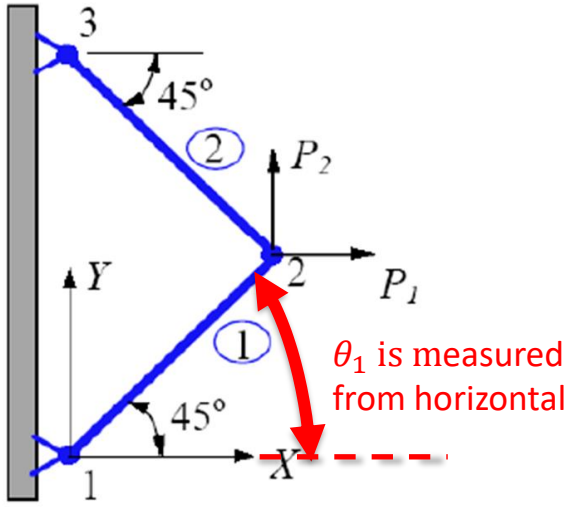
$F_1 x \quad F_1 y \quad F_2 x \quad F_2 y$

$$\frac{EA}{L}\begin{bmatrix} & & & \\ & \textbf{Stiffness matrix} & & \\ & & & \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{bmatrix} F_1 x \\ F_1 y \\ F_2 x \\ F_2 y \end{bmatrix}$$

$\theta_1 = 45^o$

$l = \cos(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$

$m = \sin(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$



$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix}$$

$l = \cos\theta$

$m = \sin\theta$

Element 1 $u_1 \quad v_1 \quad u_2 \quad v_2$

$F_1 x \quad F_1 y \quad F_2 x \quad F_2 y$

$$\frac{EA}{L}\begin{bmatrix} u_1 & v_1 & u_2 & v_2 \\ & \textbf{Stiffness matrix} & & \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{bmatrix} F_1 x \\ F_1 y \\ F_2 x \\ F_2 y \end{bmatrix}$$

$\theta_1 = 45^o$

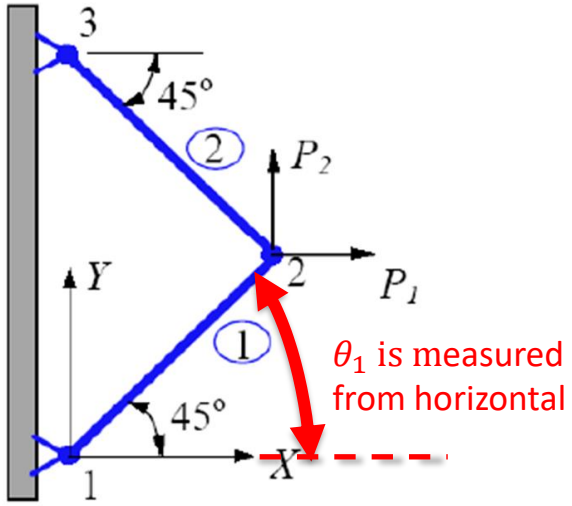
$l = \cos(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$

$m = \sin(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$

$l^2 = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5$

$m^2 = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5$

$lm = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5$



$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix} \qquad l = \cos\theta \qquad m = \sin\theta$$

Element 1 $u_1 \quad v_1 \quad u_2 \quad v_2$

$F_1 x \quad F_1 y \quad F_2 x \quad F_2 y$

$$\frac{EA}{L}\begin{bmatrix} & & & \\ & \textbf{Stiffness matrix} & & \\ & & & \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{bmatrix} F_1 x \\ F_1 y \\ F_2 x \\ F_2 y \end{bmatrix}$$

$\theta_1 = 45^o$

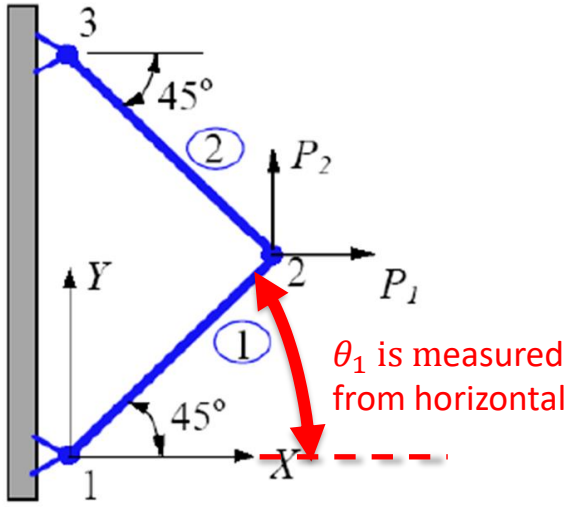
$l = \cos(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$

$m = \sin(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$

$l^2 = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5 \qquad -l^2 = -0.5$

$m^2 = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5 \qquad -m^2 = -0.5$

$lm = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5 \qquad -lm = -0.5$



$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix}\begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix}$$

$l = \cos\theta$

$m = \sin\theta$

Element 1 $u_1 \quad v_1 \quad u_2 \quad v_2$

$F_1 x \quad F_1 y \quad F_2 x \quad F_2 y$

$$\frac{EA}{L}\begin{bmatrix} & & & \\ & \textbf{Stiffness matrix} & & \\ & & & \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{bmatrix} F_1 x \\ F_1 y \\ F_2 x \\ F_2 y \end{bmatrix}$$

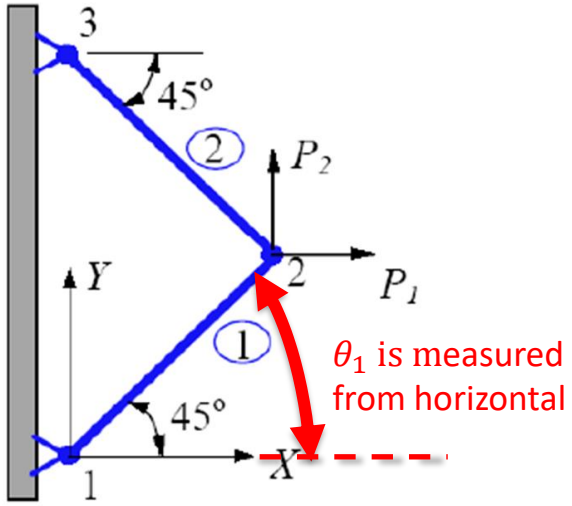
$\theta_1 = 45^o$

$l = \cos(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$

$m = \sin(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$

$l^2 = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5$	$-l^2 = -0.5$
$m^2 = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5$	$-m^2 = -0.5$
$lm = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5$	$-lm = -0.5$

Substitute these values into the formulation equation



$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix}\begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix}$$

$l = \cos\theta$

$m = \sin\theta$

Element 1 $u_1 \quad v_1 \quad u_2 \quad v_2$

$F_1 x \quad F_1 y \quad F_2 x \quad F_2 y$

$$\theta_1 = 45^o$$

$$l = \cos(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$$

$$m = \sin(45) = 0.7071 = {}^{\sqrt{2}}/_{2}$$

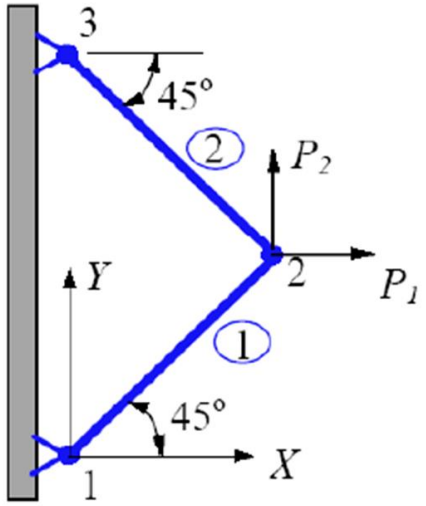
$$l^2 = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5 \qquad -l^2 = -0.5$$

$$m^2 = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5 \qquad -m^2 = -0.5$$

$$lm = \left({}^{\sqrt{2}}/_{2}\right)^2 = 0.5 \qquad -lm = -0.5$$

$$\frac{EA}{L}\begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{bmatrix} F_1 x \\ F_1 y \\ F_2 x \\ F_2 y \end{bmatrix}$$

Substitute these values into the formulation equation

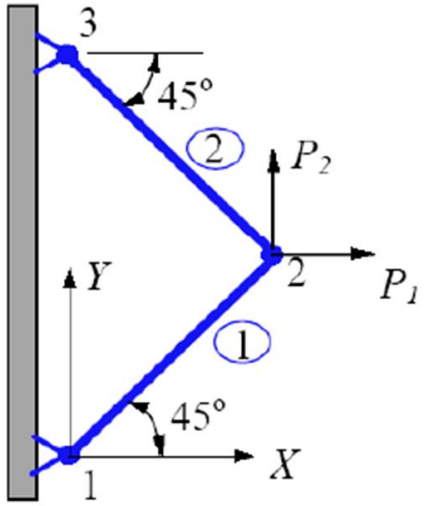


$$\begin{matrix} & u_i & v_i & u_j & v_j \\ \frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} & \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i\,x \\ F_i\,y \\ F_j\,x \\ F_j\,y \end{Bmatrix} \end{matrix}$$

$l = \cos\theta$:

$m = \sin\theta$

Element 2

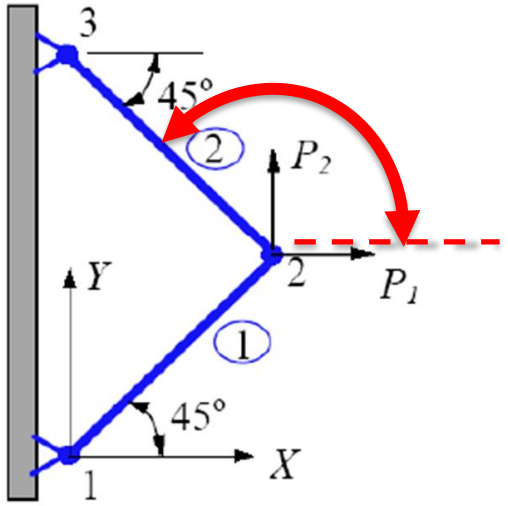


$$\begin{matrix} & u_i & v_i & u_j & v_j \\ \frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} & \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i\,x \\ F_i\,y \\ F_j\,x \\ F_j\,y \end{Bmatrix} \end{matrix} \qquad \begin{matrix} l = \cos\theta \\ \\ m = \sin\theta \end{matrix}$$

Element 2 $u_2 \quad v_2 \quad u_3 \quad v_3$

$F_2\,x \quad F_2\,y \quad F_3\,x \quad F_3\,y$

$$\begin{matrix} & u_2 & v_2 & u_3 & v_3 \\ \frac{EA}{L}\begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix} & \begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{bmatrix} F_2x \\ F_2y \\ F_3x \\ F_3y \end{bmatrix} \end{matrix}$$

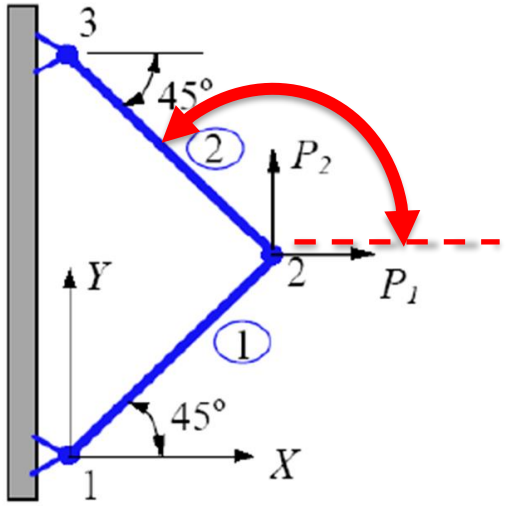


$$\frac{EA}{L}\begin{bmatrix} u_i & v_i & u_j & v_j \\ l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix} \qquad l = \cos\theta \qquad m = \sin\theta$$

Element 2 $u_2 \quad v_2 \quad u_3 \quad v_3$

$F_2 x \quad F_2 y \quad F_3 x \quad F_3 y$

$$\frac{EA}{L}\begin{bmatrix} u_2 & v_2 & u_3 & v_3 \\ & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{bmatrix} F_2 x \\ F_2 y \\ F_3 x \\ F_3 y \end{bmatrix}$$



$$\begin{matrix} & u_i & v_i & u_j & v_j \end{matrix}$$

$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix}\begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i\,x \\ F_i\,y \\ F_j\,x \\ F_j\,y \end{Bmatrix}$$

$$l = \cos\theta$$

$$m = \sin\theta$$

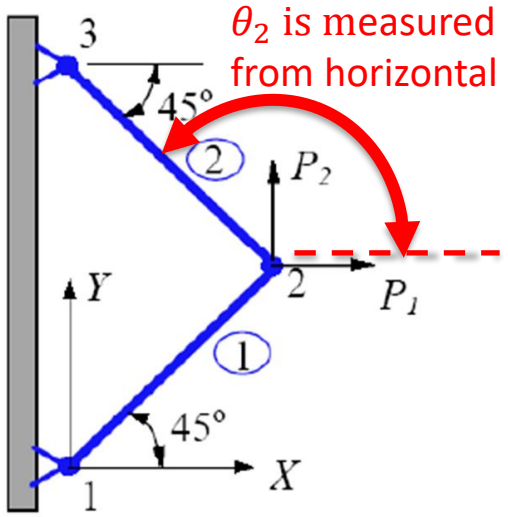
$$\begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix}$$

$$\frac{EA}{L}\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{bmatrix} F_2x \\ F_2y \\ F_3x \\ F_3y \end{bmatrix}$$

Element 2 u_2 v_2 u_3 v_3

$F_2\,x$ $F_2\,y$ $F_3\,x$ $F_3\,y$

$\theta_2 = 135^o$



θ_2 is measured from horizontal

$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix} \qquad l = \cos\theta \qquad m = \sin\theta$$

Element 2 $u_2 \quad v_2 \quad u_3 \quad v_3$

$F_2 x \quad F_2 y \quad F_3 x \quad F_3 y$

$$\frac{EA}{L}\begin{bmatrix} & u_2 & v_2 & u_3 & v_3 \\ & & & & \\ & & & & \\ & & & & \end{bmatrix} \begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{bmatrix} F_2x \\ F_2y \\ F_3x \\ F_3y \end{bmatrix}$$

$\theta_2 = 135^o$

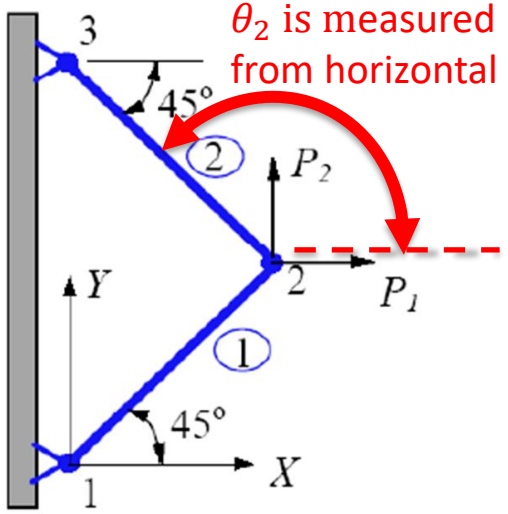
$l =$

$m =$

$l^2 =$ $\qquad -l^2 =$

$m^2 =$ $\qquad -m^2 =$

$lm =$ $\qquad -lm =$



θ_2 is measured from horizontal

$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix} \qquad l = \cos\theta \qquad m = \sin\theta$$

$$\frac{EA}{L}\begin{bmatrix} u_2 & v_2 & u_3 & v_3 \\ & & & \\ & & & \\ & & & \end{bmatrix} \begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{bmatrix} F_2x \\ F_2y \\ F_3x \\ F_3y \end{bmatrix}$$

Element 2 u_2 v_2 u_3 v_3

F_2x F_2y F_3x F_3y

$\theta_2 = 135^o$

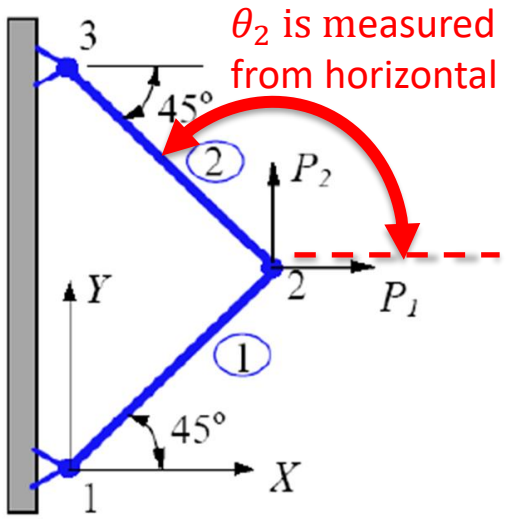
$l = \cos(135) = -0.7071 = -\sqrt{2}/2$

$m = \sin(135) = 0.7071 = \sqrt{2}/2$

$l^2 =$ $\quad -l^2 =$

$m^2 =$ $\quad -m^2 =$

$lm =$ $\quad -lm =$



$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix}\begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix} \qquad l = \cos\theta \qquad m = \sin\theta$$

$$\frac{EA}{L}\begin{bmatrix} u_2 & v_2 & u_3 & v_3 \\ & & & \\ & & & \\ & & & \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{bmatrix} F_2x \\ F_2y \\ F_3x \\ F_3y \end{bmatrix}$$

Element 2 $u_2 \quad v_2 \quad u_3 \quad v_3$

$F_2x \quad F_2y \quad F_3x \quad F_3y$

$\theta_2 = 135^o$

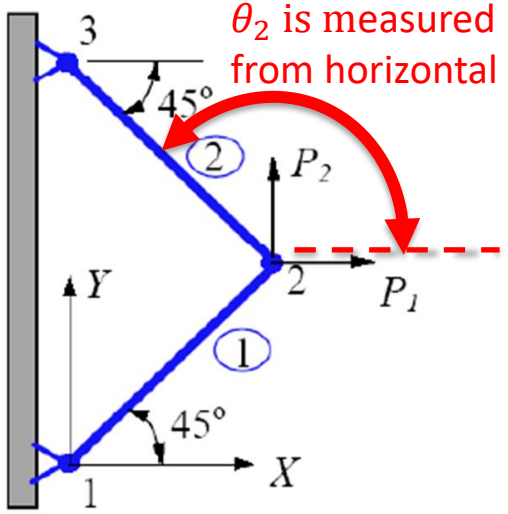
$l = \cos(135) = -0.7071 = -\sqrt{2}/2$

$m = \sin(135) = 0.7071 = \sqrt{2}/2$

$l^2 = \left(-\sqrt{2}/2\right)^2 = 0.5$ $\qquad -l^2 = -0.5$

$m^2 = \left(\sqrt{2}/2\right)^2 = 0.5$ $\qquad -m^2 = -0.5$

$lm = \left(-\sqrt{2}/2\right)\left(\sqrt{2}/2\right) = -0.5$ $\qquad -lm = 0.5$



θ_2 is measured from horizontal

$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix}\begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix} \qquad l = \cos\theta \qquad m = \sin\theta$$

$$\frac{EA}{L}\begin{bmatrix} u_2 & v_2 & u_3 & v_3 \\ & & & \\ & & & \\ & & & \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{bmatrix} F_2x \\ F_2y \\ F_3x \\ F_3y \end{bmatrix}$$

Element 2 $u_2 \quad v_2 \quad u_3 \quad v_3$

$F_2x \quad F_2y \quad F_3x \quad F_3y$

$\theta_2 = 135^o$

$l = \cos(135) = -0.7071 = -\sqrt{2}/2$

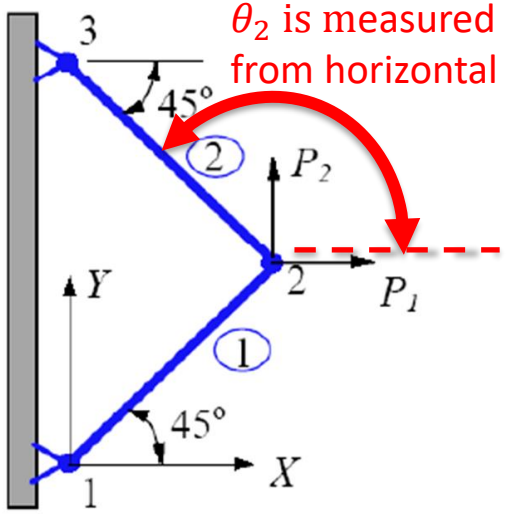
$m = \sin(135) = 0.7071 = \sqrt{2}/2$

$l^2 = \left(-\sqrt{2}/2\right)^2 = 0.5$ $\qquad -l^2 = -0.5$

$m^2 = \left(\sqrt{2}/2\right)^2 = 0.5$ $\qquad -m^2 = -0.5$

$lm = \left(-\sqrt{2}/2\right)\left(\sqrt{2}/2\right) = -0.5$ $\qquad -lm = 0.5$

Substitute these values into the formulation equation



θ_2 is measured from horizontal

$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix}\begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_i x \\ F_i y \\ F_j x \\ F_j y \end{Bmatrix} \qquad l = \cos\theta \qquad m = \sin\theta$$

$$\frac{EA}{L}\begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{bmatrix} F_2x \\ F_2y \\ F_3x \\ F_3y \end{bmatrix}$$

Element 2 u_2 v_2 u_3 v_3

F_2x F_2y F_3x F_3y

$\theta_2 = 135^o$

$l = \cos(135) = -0.7071 = -\sqrt{2}/2$

$m = \sin(135) = 0.7071 = \sqrt{2}/2$

$l^2 = \left(-\sqrt{2}/2\right)^2 = 0.5$ $\quad -l^2 = -0.5$

$m^2 = \left(\sqrt{2}/2\right)^2 = 0.5$ $\quad -m^2 = -0.5$

$lm = \left(-\sqrt{2}/2\right)\left(\sqrt{2}/2\right) = -0.5$ $\quad -lm = 0.5$

Substitute these values into the formulation equation

Element 1

$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \end{matrix}\begin{bmatrix} u_1 & v_1 & u_2 & v_2 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

$$\frac{EA}{L}\begin{matrix} \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{bmatrix} u_2 & v_2 & u_3 & v_3 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Element 1

$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \end{matrix}\begin{bmatrix} u_1 & v_1 & u_2 & v_2 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

$$\frac{EA}{L}\begin{matrix} \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{bmatrix} u_2 & v_2 & u_3 & v_3 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Need to know the total no of dof?

Element 1

$$\frac{EA}{L}\begin{matrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{matrix}\begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

$$\frac{EA}{L}\begin{matrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Need to know the total no of dof?

$dof_T = (No.\,of\ nodes)\ x\ (dof\ each\ node)$

$dof_T = \mathbf{3}\ nodes\ x\ \mathbf{2}\ dof = 6$

Element 1

$$\frac{EA}{L}\begin{bmatrix} & u_1 & v_1 & u_2 & v_2 \\ & 0.5 & 0.5 & -0.5 & -0.5 \\ & 0.5 & 0.5 & -0.5 & -0.5 \\ & -0.5 & -0.5 & 0.5 & 0.5 \\ & -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

$$\frac{EA}{L}\begin{bmatrix} & u_2 & v_2 & u_3 & v_3 \\ & 0.5 & -0.5 & -0.5 & 0.5 \\ & -0.5 & 0.5 & 0.5 & -0.5 \\ & -0.5 & 0.5 & 0.5 & -0.5 \\ & 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Need to know the total no of dof?

$dof_T = (No.\,of\ nodes)\ x\ (dof\ each\ node)$

$dof_T = \mathbf{3}\ nodes\ x\ \mathbf{2}\ dof = 6$

$Global\ stiffness\ matrix\ size = 6\ x\ 6\ matrix$

Element 1

$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

$$\frac{EA}{L}\begin{matrix} \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

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$dof_T = \mathbf{3}\ nodes\ x\ \mathbf{2}\ dof = 6$

$Global\ stiffness\ matrix\ size = 6\ x\ 6\ matrix$

$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Element 1

$$\frac{EA}{L}\begin{matrix} u_1 & v_1 & u_2 & v_2 \end{matrix}\begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

$$\frac{EA}{L}\begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix}\begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Need to know the total no of dof?

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$dof_T = \mathbf{3}\ nodes\ x\ \mathbf{2}\ dof = 6$

$Global\ stiffness\ matrix\ size = 6\ x\ 6\ matrix$

$$\frac{EA}{L}\begin{matrix} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 \end{matrix}\begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 & & \\ 0.5 & 0.5 & -0.5 & -0.5 & & \\ -0.5 & -0.5 & 0.5 & 0.5 & & \\ -0.5 & -0.5 & 0.5 & 0.5 & & \\ & & & & & \\ & & & & & \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Element 1

$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

$$\frac{EA}{L}\begin{matrix} \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

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$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 & & \\ 0.5 & 0.5 & -0.5 & -0.5 & & \\ -0.5 & -0.5 & 1 & 0 & -0.5 & 0.5 \\ -0.5 & -0.5 & 0 & 1 & 0.5 & -0.5 \\ & & -0.5 & 0.5 & 0.5 & -0.5 \\ & & 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Element 1

$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

$$\frac{EA}{L}\begin{matrix} \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

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$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 & 0 & 0 \\ 0.5 & 0.5 & -0.5 & -0.5 & 0 & 0 \\ -0.5 & -0.5 & 1 & 0 & -0.5 & 0.5 \\ -0.5 & -0.5 & 0 & 1 & 0.5 & -0.5 \\ 0 & 0 & -0.5 & 0.5 & 0.5 & -0.5 \\ 0 & 0 & 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Element 1

$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

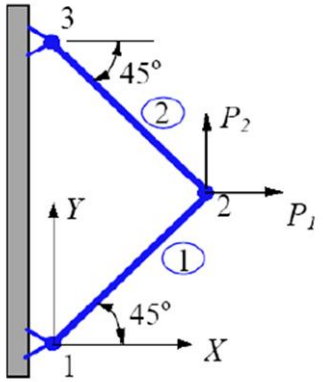
$$\frac{EA}{L}\begin{matrix} \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Need to know the total no of dof?

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$dof_T = \mathbf{3}\ nodes\ x\ \mathbf{2}\ dof = 6$

$Global\ stiffness\ matrix\ size = 6\ x\ 6\ matrix$



$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 & 0 & 0 \\ 0.5 & 0.5 & -0.5 & -0.5 & 0 & 0 \\ -0.5 & -0.5 & 1 & 0 & -0.5 & 0.5 \\ -0.5 & -0.5 & 0 & 1 & 0.5 & -0.5 \\ 0 & 0 & -0.5 & 0.5 & 0.5 & -0.5 \\ 0 & 0 & 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Element 1

$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

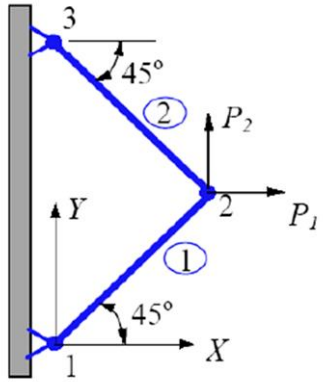
$$\frac{EA}{L}\begin{matrix} \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Need to know the total no of dof?

$dof_T = (No.\,of\ nodes)\ x\ (dof\ each\ node)$

$dof_T = \mathbf{3}\ nodes\ x\ \mathbf{2}\ dof = 6$

$Global\ stiffness\ matrix\ size = 6\ x\ 6\ matrix$



$$\frac{EA}{L}\begin{matrix} \\ u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}\begin{matrix} \begin{matrix} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 & 0 & 0 \\ 0.5 & 0.5 & -0.5 & -0.5 & 0 & 0 \\ -0.5 & -0.5 & 1 & 0 & -0.5 & 0.5 \\ -0.5 & -0.5 & 0 & 1 & 0.5 & -0.5 \\ 0 & 0 & -0.5 & -0.5 & 0.5 & 0.5 \\ 0 & 0 & -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix} \end{matrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Element 1

$$\frac{EA}{L}\begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

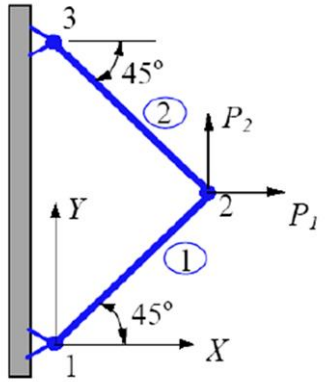
$$\frac{EA}{L}\begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Need to know the total no of dof?

$dof_T = (No.\,of\ nodes)\ x\ (dof\ each\ node)$

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$Global\ stiffness\ matrix\ size = 6\ x\ 6\ matrix$



$$\frac{EA}{L}\begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 & 0 & 0 \\ 0.5 & 0.5 & -0.5 & -0.5 & 0 & 0 \\ -0.5 & -0.5 & 1 & 0 & -0.5 & 0.5 \\ -0.5 & -0.5 & 0 & 1 & 0.5 & -0.5 \\ 0 & 0 & -0.5 & -0.5 & 0.5 & 0.5 \\ 0 & 0 & -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

$$\frac{EA}{L}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \end{Bmatrix}$$

Element 1

$$\frac{EA}{L}\begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \end{Bmatrix}$$

Element 2

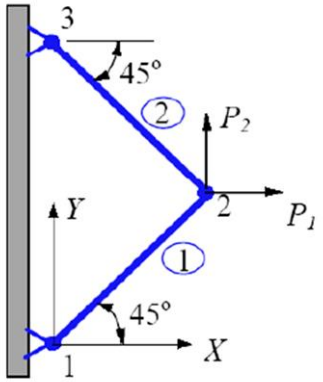
$$\frac{EA}{L}\begin{bmatrix} 0.5 & -0.5 & -0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

Need to know the total no of dof?

$dof_T = (No.\,of\ nodes)\ x\ (dof\ each\ node)$

$dof_T = \mathbf{3}\ nodes\ x\ \mathbf{2}\ dof = 6$

$Global\ stiffness\ matrix\ size = 6\ x\ 6\ matrix$

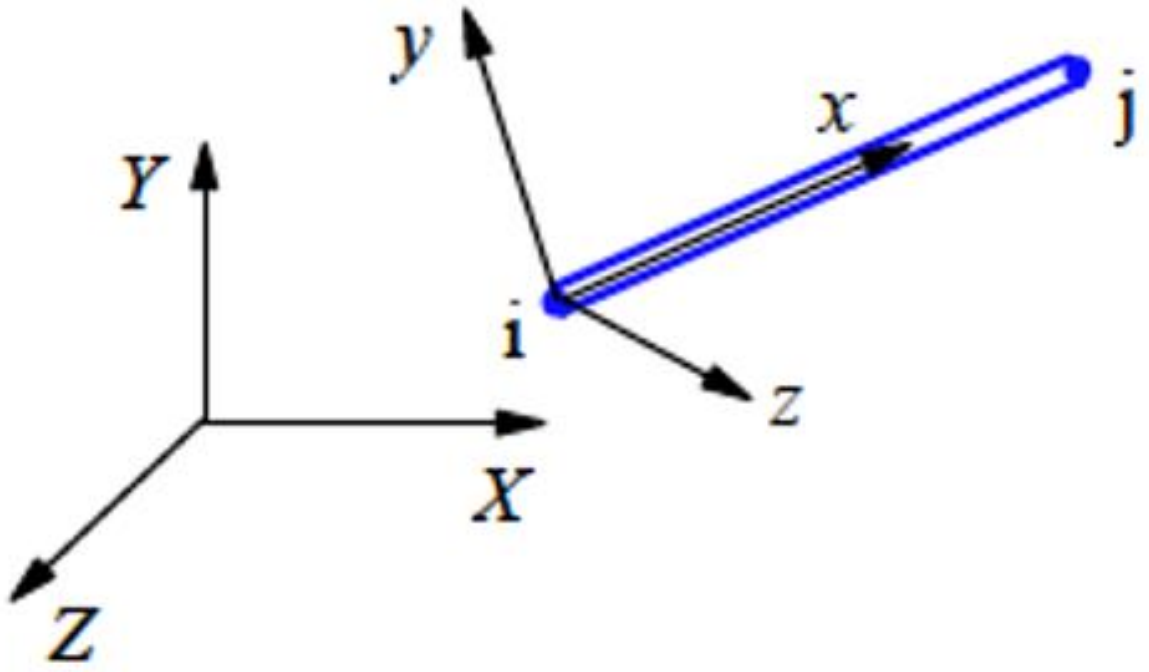


$$\frac{EA}{L}\begin{bmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 1 & 0 & -0.5 & 0.5 \\ \cdot & \cdot & 0 & 1 & 0.5 & -0.5 \\ \cdot & \cdot & -0.5 & 0.5 & 0.5 & -0.5 \\ \cdot & \cdot & 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_1x \\ F_1y \\ P_1 \\ P_2 \\ F_3x \\ F_3y \end{Bmatrix}$$

$$\begin{Bmatrix} u_2 \\ v_2 \end{Bmatrix} = \frac{L}{EA}\begin{Bmatrix} P_1 \\ P_2 \end{Bmatrix}$$

$$\frac{EA}{L}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{Bmatrix} u_2 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} P_1 \\ P_2 \end{Bmatrix}$$

3D- Truss Element



Each node can deform in 3 axes (u, v, w)

Nodal forces on 3 axes (F_x, F_y, F_z)

3D- Truss Element

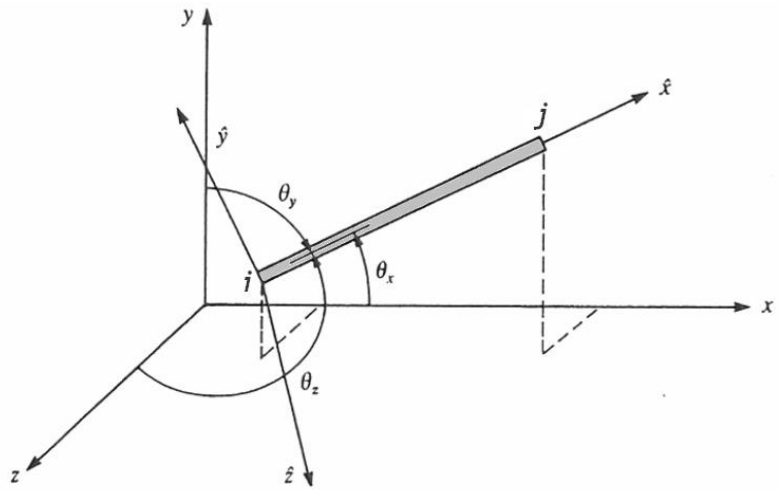
$$
\begin{array}{c}
\\
\frac{EA}{L}
\end{array}
\begin{array}{c}
\begin{array}{cccccc} \mu_i & \nu_i & \omega_i & \mu_j & \nu_j & \omega_j \end{array} \\
\begin{bmatrix}
l_1^2 & l_1m_1 & l_1n_1 & -l_1^2 & -l_1m_1 & -l_1n_1 \\
l_1m_1 & m_1^2 & m_1n_1 & -l_1m_1 & -m_1^2 & -m_1n_1 \\
l_1n_1 & m_1n_1 & n_1^2 & l_1n_1 & m_1n_1 & -n_1^2 \\
-l_1^2 & -l_1m_1 & -l_1n_1 & l_1^2 & l_1m_1 & l_1n_1 \\
-l_1m_1 & -m_1^2 & -m_1n_1 & l_1m_1 & m_1^2 & m_1n_1 \\
-l_1n_1 & -m_1n_1 & -n_1^2 & l_1n_1 & m_1n_1 & n_1^2
\end{bmatrix}
\end{array}
\begin{Bmatrix} \mu_i \\ \nu_i \\ \omega_i \\ \mu_j \\ \nu_j \\ \omega_j \end{Bmatrix}
= \begin{bmatrix} F_ix \\ F_iy \\ F_iz \\ F_jx \\ F_jy \\ F_jz \end{bmatrix}
$$

This is for one element

3D- Truss Element

$$
\begin{array}{cc}
 & \begin{matrix} \mu_i & \nu_i & \omega_i & \mu_j & \nu_j & \omega_j \end{matrix} \\
\dfrac{EA}{L} & \begin{bmatrix}
l_1^2 & l_1m_1 & l_1n_1 & -l_1^2 & -l_1m_1 & -l_1n_1 \\
l_1m_1 & m_1^2 & m_1n_1 & -l_1m_1 & -m_1^2 & -m_1n_1 \\
l_1n_1 & m_1n_1 & n_1^2 & l_1n_1 & m_1n_1 & -n_1^2 \\
-l_1^2 & -l_1m_1 & -l_1n_1 & l_1^2 & l_1m_1 & l_1n_1 \\
-l_1m_1 & -m_1^2 & -m_1n_1 & l_1m_1 & m_1^2 & m_1n_1 \\
-l_1n_1 & -m_1n_1 & -n_1^2 & l_1n_1 & m_1n_1 & n_1^2
\end{bmatrix}
\end{array}
\begin{Bmatrix} \mu_i \\ \nu_i \\ \omega_i \\ \mu_j \\ \nu_j \\ \omega_j \end{Bmatrix}
=
\begin{bmatrix} F_ix \\ F_iy \\ F_iz \\ F_jx \\ F_jy \\ F_jz \end{bmatrix}
$$

This is for one element



$$l = cos\theta_x$$
$$m = cos\theta_y$$
$$n = cos\theta_z$$

Exercise: Effect of Node Connection Direction on Global Reference Frame Stiffness Matrix

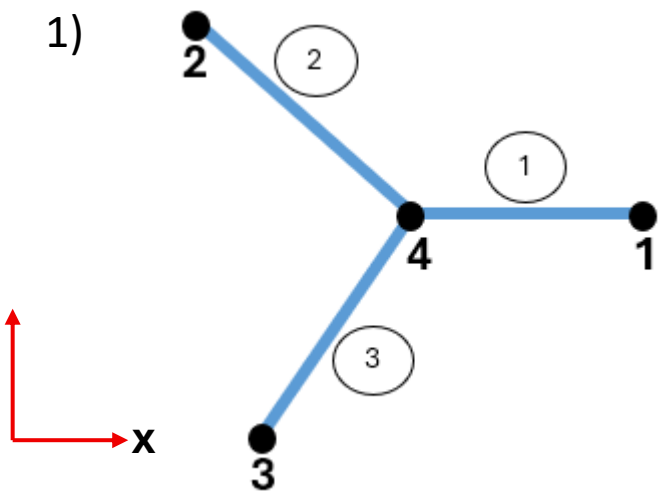
For each structure**, calculate the stiffness matrix in global reference frame** of every element using the node connection in both directions. Compare the two matrices for the same physical element. A = 2.00 x 10^{11}N/m^2, A = 5x 10^{-4} m and L = 1.00 m. **Obtain the global assembly matrix for both structures.**

$$\frac{EA}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{Bmatrix} = \begin{Bmatrix} F_{i\,x} \\ F_{i\,y} \\ F_{j\,x} \\ F_{j\,y} \end{Bmatrix}$$

$l = \cos\theta$;

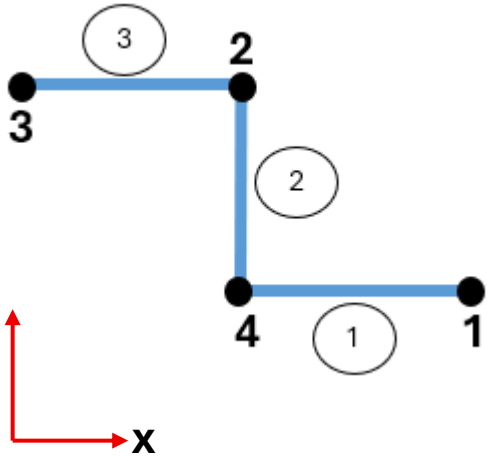
$m = \sin\theta$

1)



Element	Direction 1	Angle	Direction2	Angle
1	1-4	180	4-1	0
2	4-2	135	2-4	315
3	4-3	225	3-4	45

2)



Element	Direction 1	Angle	Direction2	Angle
1	1-4	180	4-1	0
2	4-2	90	2-4	270
3	3-2	180	3-4	45

Practice 2 - Truss Element

Find the horizontal displacement of node 1.

